Embeddings and neighbourhoods

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

Make row mapping,table version and candidate policy visible.

Use complete vectors and exact tie examples;separate constructed geometry,mask-aware averages and an artificial fitted-row objective.

Keep one habit: Assess independent distances,ties and mean-order collision with checked keys and the correct candidate roles.

MAIN LESSON · SECTION 1

Prepare the connected mathematical investigation

> Keep explanation,shared practice and independent evidence distinct.

Target and main teaching plan

Primary target: Look up a vector using a token ID and preserve sequence shape

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min

> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together and Build your understanding; use the matching workbook W2 and Individual reasoning check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Preparation and prerequisite

Printed lesson and workbook; paper and pencil

Calculator and optional local Python 3 editor; no packages or AI accounts needed

Use the fictional examples supplied; prepare the fresh retry before class

IDs are assigned identifiers. Their numerical order and spacing have no declared semantic meaning; a consistent relabelling can change those differences without changing the symbols.

Opening retrieval

Without opening the old lesson, explain one key idea from Sequence representations. Give an example and a mistake that the idea helps you avoid.

Look for: Encode an ordered sequence with an explicit vocabulary

Reference example, not the only acceptable wording: Teacher explanation: IDs are [2, 3, 4, 0], mask [1, 1, 1, 0], real length 3 and final real event STOP at position 2.

Without opening the old lesson, explain one key idea from **Training** a small image model. Give an example and a mistake that the idea helps you avoid.

Look for: Trace a complete image model with fixed features and one trained weight

Reference example, not the only acceptable wording: Teacher explanation: d = 2, h1 = 2 and h2 = 0. Score is 0.9, **prediction** class 1 and squared error (0.9 - 1) squared = 0.01.

Without opening the old lesson, explain one key idea from Image tensors and channels. Give an example and a mistake that the idea helps you avoid.

Look for: Read a pixel using declared row column and channel axes

Reference example, not the only acceptable wording: Teacher explanation: The RGB pixel is [0, 0, 255]; its blue value is 255. Shape is 2 by 2 by 3, with four pixels and twelve channel values.

Pictured end goal

By the end, I can explain embeddings and **neighbourhood**s using a traced example and a stated limitation.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Sequence representations

Training a small image model

Image tensors and channels

For the compact alternative schedule, use two 45-minute sessions. Begin with the difference between an identifier and its looked-up vector.

The original table is authored, not learned; the separate one-row update demonstrates optimisation under a stated artificial objective. Do not attach invented semantic labels to the two coordinate axes.

Keep original and updated table versions separate.

Outcomes

Look up a vector using a token ID and preserve sequence shape

Calculate nearest neighbours with an explicit distance and tie rule

Distinguish constructed vectors from learned features and limit similarity claims

Preparation

Prerequisite answer

End goal for the pictorial case

This added worked case illustrates the week’s main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

Teach geometry and row meaning with an explicit candidate policy

> Keep original vocabulary geometry,pictured P/Q/R and independent A/B/Ctables separate.

Main lookup definitions,complete distances and guided key

Continue the fictional robot vocabulary and add WAIT as ID 5. Use the table below, chosen by the lesson author for arithmetic practice. These coordinates are not measured robot properties and were not learned from data.

Each token ID selects one row; the resulting two-component vector is its embedding in this example.

The table shape is 6 by 2: six vocabulary entries and **embedding dimension** 2. The ID sequence [2, 4, 0, 0] becomes [[2, 0], [0, 2], [0, 0], [0, 0]], with shape 4 by 2. Its mask remains [1, 1, 0, 0].

The vector dimension is not the sequence length or vocabulary size.

A zero PAD vector does not remove the need for masking. Averaging all four vectors above gives [0.5, 0.5]; averaging only the two real vectors gives [1, 1].

UNK is a real position when it replaces an unknown symbol, so its vector is not automatically excluded.

Use **squared Euclidean distance**: square each component difference and add. From GO [2, 0] to TURN [1, 1], the distance is (2 - 1) squared + (0 - 1) squared = 2. To STOP it is 8; to WAIT it is 5.

TURN is therefore the nearest of those three under this table and metric.

The candidate set excludes PAD, UNK and the query token itself. Otherwise a query would normally retrieve itself at distance zero.

This code assumes finite numeric components and uses exact ties, appropriate for these integer examples. Floating-point comparisons in other experiments need a declared precision policy.

Candidate in original table	Vector	Squared distance from GO[2, 0]	Squared distance from query[1, 0]
GO	[2, 0]	Excluded self	1
TURN	[1, 1]	2	1
STOP	[0, 2]	8	5
WAIT	[0, 1]	5	2
PAD /UNK	[0, 0]/[–1, –1]	Excluded reserved roles	Excluded reserved roles

For a named-token query GO, exclude GO itself as well as reserved PAD/UNKroles. For the arbitrary query[1, 0], use all four regular candidates; it is not a named token in the table. nearest returns every exact nearest tie, sorted only for display. These integer coordinates make exact ties checkable; other numerical experiments need a declared precision policy.

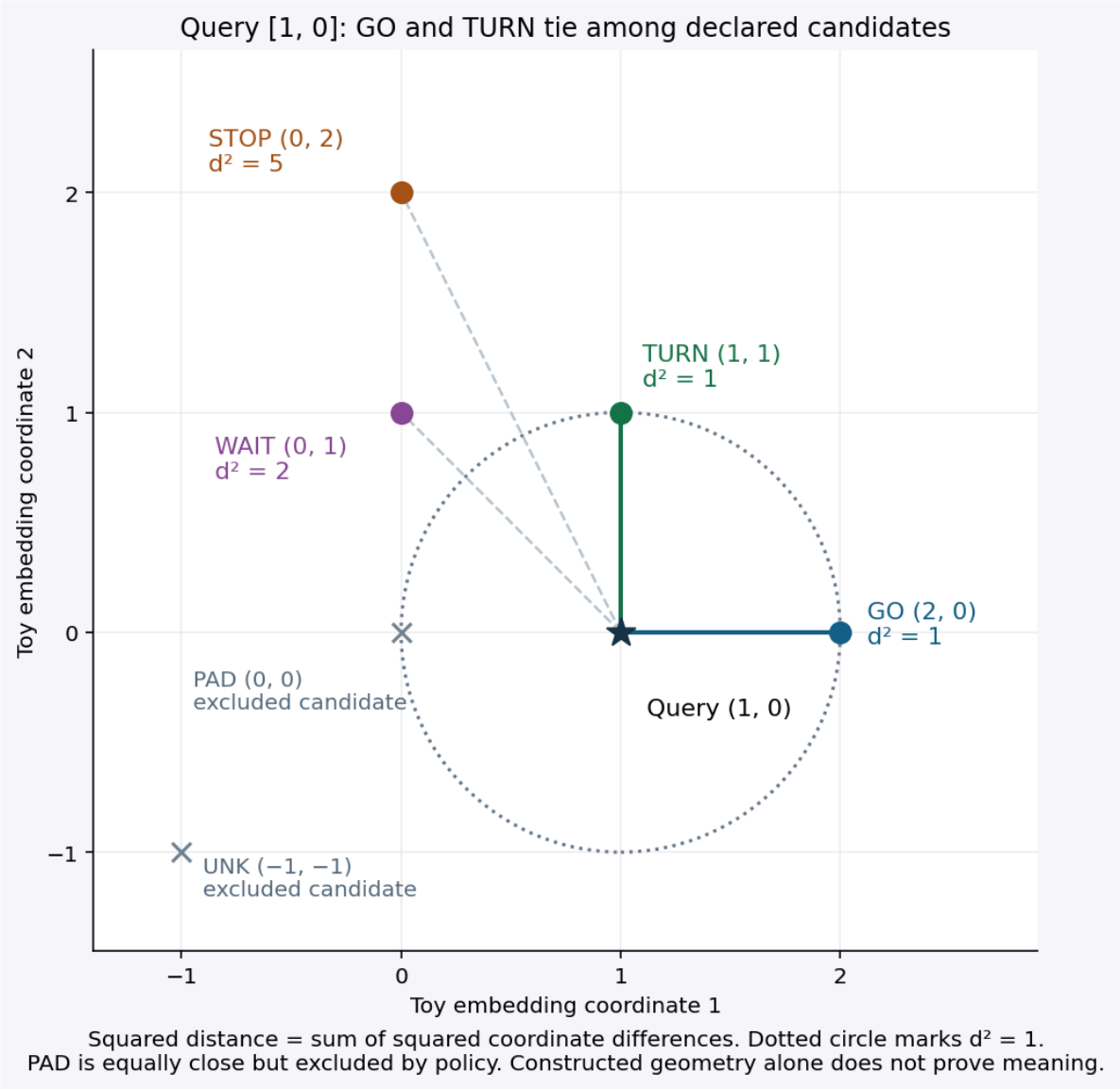
The functions check nonempty/equal dimensions and nonempty candidates but assume finite numeric components. All coordinates in this original table were hand-chosen.

Guided: STOP distances to GO, TURN and WAIT are 8, 2 and 1, so WAIT is nearest. Exit: the original table and squared Euclidean rule place TURN nearest to GO among the stated candidates.

That is not proof of equivalent meaning or substitutability.

A lower artificial **training** objective only supports improvement on that objective; useful task behaviour requires appropriate independent evaluation and error analysis.

Labelled tie and isolated fitted-row evidence



Separate isolated GO-row fitting example	Complete value / status
Original GO row and chosen target	e[2, 0], target[0, 2]; target author-supplied for arithmetic
Artificial objective	$J(e)=(e1-0)^2+(e2-2)^2$
Gradient and rate	[4, −4], rate.25
New row in a copied table	[1, 1]; every other row remains frozen

Separate isolated GO-row fitting example	Complete value / status
Objective before /after	8→2
Collision with TURN	Both rows now[1, 1]; a vector-only deterministic head cannot distinguish them
Evidence limit	One defined objective decreased; useful semantic neighbourhoods were not established

Separate pictured table and query scope

Measure embedding neighbours without turning numerical proximity into factual certainty.

Toy table P =[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use **squared Euclidean distance**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

The picture is a separate three-vector table:P[0, 1], Q[3, 1], R[1, 4]. Query[1, 1]has squared distances 1, 4, 9, so P is nearest. Move only the query to[2, 1]:distances 4, 1, 10, so Q is nearest. The original GO/TURN/STOP/WAITtable is unchanged and uses different coordinates/names. Both examples establish declared geometry, not factual truth, topic identity or a useful substitution rule.

Trace the changed query without a semantic overclaim

AI ACADEMY / AI-5/26 · 02 · TRACE THE CASE

Follow the deciding steps

01 Distance to P

$(1-0)^2+(1-1)^2=1$.

02 Distances to Q/R

Q distance 4; R distance 9.

03 Select neighbour

P is nearest under the declared metric; this is a geometric result, not proof of meaning.

Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Distance to P	$(1-0)^2+(1-1)^2=1$.
2. Distances to Q/R	Q distance 4; R distance 9.
3. Select neighbour	P is nearest under the declared metric; this is a geometric result, not proof of meaning.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

An embedding stores each item as an ordered vector. Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

Embedding geometry depends on **training**, metric and scale; proximity does not prove truth, causation or source permission.

Change: Move the query to [2, 1].

Prediction: Q becomes nearest with squared distance 1; P distance 4; R distance 10.

Reason: Changing the query position changes geometric neighbours while the stored table remains fixed.

▼ Earlier wording, alternatives and picture transcripts

Term	Meaning
Embedding table	A table assigning a fixed-length numerical vector to each token ID
Embedding dimension	The number of components in each embedding vector
Neighbourhood	Nearby candidates under a stated representation and distance rule
Squared Euclidean distance	The sum of squared component differences between two vectors

One query can have two closest candidates

CANDIDATE VECTOR	QUERY VECTOR	SQUARED DISTANCE
GO [2, 0]	[1, 0]	1
TURN [1, 1]	[1, 0]	1
WAIT [0, 1]	[1, 0]	2
STOP [0, 2]	[1, 0]	5

The nearest set is GO and TURN. Alphabetical display order does not break the mathematical tie or make one candidate more similar.

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

One query can have two closest candidates

CANDIDATE VECTOR	QUERY VECTOR	SQUARED DISTANCE
GO [2, 0]	[1, 0]	1
TURN [1, 1]	[1, 0]	1
WAIT [0, 1]	[1, 0]	2
STOP [0, 2]	[1, 0]	5

The nearest set is GO and TURN. Alphabetical display order does not break the mathematical tie or make one candidate more similar.

Using STOP [0, 2] as the query, calculate distances to GO, TURN and WAIT. Exclude STOP itself and both reserved entries. Which candidate is nearest?

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.

Thinking move	Teacher prompt
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

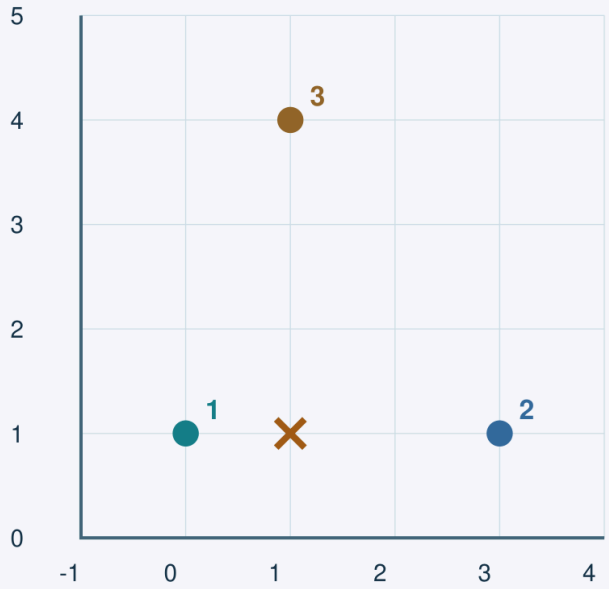
Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

A pictorial worked case: Embeddings and neighbourhoods

The facts we will use

AI ACADEMY / AI-5/26

Squared Euclidean embedding distances



Horizontal: first embedding coordinate; vertical: second embedding coordinate

1: P at (0, 1)

2: Q at (3, 1)

3: R at (1, 4)

Cross: query at (1, 1)

Distance rule: Squared Euclidean distance

Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/26

Embeddings and neighbourhoods

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.

1. Distance to P

(1-0)2+(1-1)2=1.

2. Distances to Q/R

Q distance 4; R distance 9.

3. Select neighbour

P is nearest under the declared metric; this is a geometric result, not proof of meaning.

CHANGE ONE THING

Move the query to [2, 1].

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

MAIN LESSON · SECTION 3

Model a step,then remove help

> Collect each learner’s reasoning before showing a key.

Current completion task and checked reasoning

Completion task: For query [1, 0], compute squared distances to GO, TURN, STOP and WAIT in the original table. Return all nearest candidates.

Teacher check: Teacher explanation: Distances are 1, 1, 5 and 2 respectively. The nearest set is GO and TURN; ordering their names does not remove the tie.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Individual misconception check

Use the worked case for Embeddings and **neighbourhood**s. Proposed explanation: “Token ID differences define embedding distances”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: The ID selects a row; distances are computed between row vectors.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictured reasoning and changed case

What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Evidence about how the table was learned plus task-specific evaluation. These invented coordinates alone establish only distances. Changed-case check: Q becomes nearest with squared distance 1; P distance 4; R distance 10.

Changing the query position changes geometric neighbours while the stored table remains fixed.

Changed-case reasoning: Q becomes nearest with squared distance 1; P distance 4; R distance 10. Changing the query position changes geometric neighbours while the stored table remains fixed.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Accessible support and extension

Use See how it works and Follow a worked example. Collect a **prediction** before revealing its result; ask the learner to point to each intermediate fact.

Provide separate columns for token, ID and vector. Use a distance ledger with two differences, two squares and a sum; allow calculators.

Extension: consistently permute all table row IDs and show that token-level neighbour results stay unchanged, then deliberately omit the table permutation to expose the mismatch.

▼ Earlier wording, alternatives and picture transcripts

Time or stage	Teacher action
Session 1 start to 7 minutes	Retrieve arbitrary IDs and label vocabulary size, sequence length and vector dimension.
7 to 18 minutes	Trace table lookups and masks, then complete W1.
18 to 32 minutes	Compute GO distances by hand and run or trace the nearest-neighbour code.
32 to 45 minutes	Use W2 and the visual to preserve ties under an explicit candidate policy.
Session 2 start to 8 minutes	Complete the guided STOP query before changing any table row.
8 to 20 minutes	Trace the isolated learning update and complete W3, noting the resulting collision.
20 to 35 minutes	Collect W4 independently, requiring distances, the full nearest set and an order-loss explanation.
35 to 45 minutes	Use W5 and the exit response to audit interpretation and version consistency, then administer the fresh retry.

Reduce help deliberately

Individual reasoning check

Pictorial practice question

Reasoned teacher answer

MAIN LESSON · SECTION 4

Run the complete supplied investigation

> Separate predictions,actual runs and paper traces.

Complete inputs,expected results and limits

W1TURN WAIT STOPplusPADgives vectors[[1, 1], [0, 1], [0, 2], [0, 0]], mask[1, 1, 1, 0], shape 4×2. W2 query[1, 0]distances 1/1/5/2 forGO/TURN/STOP/WAIT, bothGO/TURN. W3 isolatedGOgradient[4, −4], new[1, 1], loss 8→2, TURNcollision; no useful semantics proved. W5 swappingIDsalonechangesselectedrows; save/change vocabulary mapping, table version and consumers consistently.

IndependentA[0, 0], B[2, 0], C[1, 2], query[1, 0]distances 1/1/4, bothA/B; meanA/B[1, 0]and reversemeanidentical. Candidate A is valid in that separate table, not a PAD role inferred from zero coordinates.

Diagnose the current misconceptions

Token ID differences define embedding distances: The ID selects a row; distances are computed between row vectors.

Nearest always means interchangeable: Closeness depends on representation, metric and candidates, and need not preserve task labels or meaning.

A learned embedding cannot lose identity: Two rows can coincide, and sequence aggregation can create additional collisions.

Use Run the investigation for the **evidence** ledger. Add a boundary or missing-**input** case and record the expected response before checking. More successful repeats of one case do not establish wider coverage.

Choose accessible support

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ **Earlier wording, alternatives and picture transcripts**

Match the support to the first error

MAIN LESSON · SECTION 5

Assess the new schema or calculation transfer

> Keep the selected answer private until after a first attempt.

Selected independent task and checked answer

Selected case: A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query.

Can that mean identify which ordered pair produced it?

Teacher explanation: Distances to A, B and C are 1, 1 and 4. A and B tie. Their mean is [1, 0], equal to the query. Reversing A B to B A gives the same mean, so the mean alone cannot identify the order.

Judge explanation,trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Trace the lookup shape

Using the original table, embed TURN WAIT STOP with one right-padding slot. Give the vector sequence, mask and shape.

Teacher explanation: Vectors are [[1, 1], [0, 1], [0, 2], [0, 0]], mask [1, 1, 1, 0] and shape 4 by 2.

W2 Calculate distances and a tie

For query [1, 0], compute squared distances to GO, TURN, STOP and WAIT in the original table. Return all nearest candidates.

Teacher explanation: Distances are 1, 1, 5 and 2 respectively. The nearest set is GO and TURN; ordering their names does not remove the tie.

W3 Trace a trainable row

For the isolated GO-row update, give the gradient, updated vector, starting loss and new loss. Which other token now shares the vector?

Teacher explanation: Gradient [4, -4], updated vector [1, 1], loss 8 before and 2 after. TURN already has [1, 1]. This collision does not establish equivalent meaning.

W5 Keep the lookup consistent

A developer swaps the IDs of GO and STOP but leaves the embedding rows unchanged. What goes wrong, and what must be saved or changed together?

Teacher explanation: Each token now selects the other token's old row, changing the encoded **input**. Save and update the vocabulary-to-row mapping and table together, along with version and downstream model.

ID relabelling is harmless only when every lookup and consumer stays consistent.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Guided and exit questions

Guided: STOP distances to GO, TURN and WAIT are 8, 2 and 1, so WAIT is nearest. Exit: the original table and squared Euclidean rule place TURN nearest to GO among the stated candidates.

That is not proof of equivalent meaning or substitutability.

A lower artificial **training** objective only supports improvement on that objective; useful task behaviour requires appropriate independent evaluation and error analysis.

W1 Trace the lookup shape

W2 Calculate distances and a tie

W3 Trace a trainable row

W4 Find a neighbourhood independently

A separate table contains A [0, 0], B [2, 0] and C [1, 2]. For query [1, 0], give all squared distances and every nearest candidate. Then calculate the mean of A and B and compare it with the query.
</details>

Can that mean identify which ordered pair produced it?

Teacher explanation: Distances to A, B and C are 1, 1 and 4. A and B tie. Their mean is [1, 0], equal to the query. Reversing A B to B A gives the same mean, so the mean alone cannot identify the order.

W5 Keep the lookup consistent

Independent assessment and fresh retry

W4 requires distances 1, 1, 4, both A and B, mean [1, 0] and a reversed-order collision. Fresh retry: query [0, 1], candidates D [0, 0], E [0, 2], F [3, 1]. Distances are 1, 1, 9; D and E tie.

Their mean is [0, 1], and swapping their order leaves it unchanged.

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect,support and revisit the mathematics

> Use the saved evidence to choose the next step.

Project and next-week bridge

Add a labelled embeddings and **neighbourhood**s **evidence** sheet to Image-pattern Network Investigation. Show how you can look up a vector using a token id and preserve sequence shape.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

Provide separate columns for token, ID and vector. Use a distance ledger with two differences, two squares and a sum; allow calculators.

Extension: consistently permute all table row IDs and show that token-level neighbour results stay unchanged, then deliberately omit the table permutation to expose the mismatch.

Week 27 explores similarity and normalisation, including changes in neighbour ranking.

Retain the project ledger and assess a changed case after support. Check the relevant prerequisite before carrying this component forward.

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding.

Delayed return and support record

Suggested return points in this level: ai-5/27, ai-5/29, ai-5/32. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

▼ Earlier wording, alternatives and picture transcripts

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Alternative planning checklist**

REFERENCE · SECTION 7

Reference · Alternative planning checklist

> Use alongside the main route when helpful.

Planning,assessment and project reminders

Teaching target: Look up a vector using a token ID and preserve sequence shape

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. Check a local Python 3 environment before class; have a paper-trace version ready.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Words that unlock the idea and Follow a worked example; record recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Plan:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
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> **Plan:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project and Try a new case independently. Allow extra time for the full task set. When code is supplied, use the examples in See how it works or Follow a worked example and the browser lab in Supported practice, repair and portfolio.

Before class, solve the supported practice: For query [1, 0], compute squared distances to GO, TURN, STOP and WAIT in the original table. Return all nearest candidates.

Reasoned practice answer: Teacher explanation: Distances are 1, 1, 5 and 2 respectively. The nearest set is GO and TURN; ordering their names does not remove the tie.

Check every learner before discussion: Use the worked case for Embeddings and **neighbourhood**s. Proposed explanation: “Token ID differences define embedding distances”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: The ID selects a row; distances are computed between row vectors.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Move the query to [2, 1]. Require a **prediction** and a reason before comparing. Expected result: Q becomes nearest with squared distance 1; P distance 4; R distance 10.

Reason: Changing the query position changes geometric neighbours while the stored table remains fixed.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Prepare the demonstration. Gather reproducible instructions, **evidence**, access checks and remaining limitations.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference · Feedback,labs and repair paths**

REFERENCE · SECTION 8

Reference · Feedback,labs and repair paths

> Retain the first attempt and the evidence used for a repair.

Feedback and record guidance

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

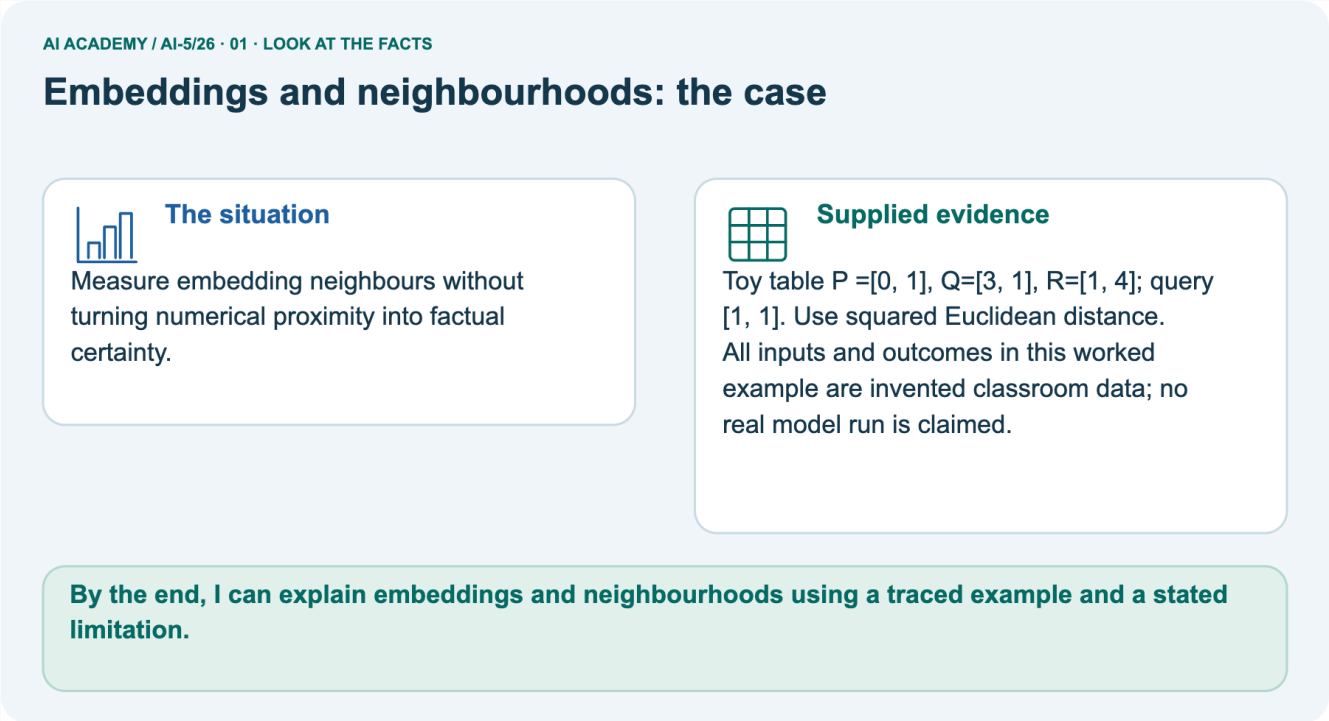
▼ **Reference · Teach with the six pictures**

REFERENCE · SECTION 9

Reference · Teach with the six pictures

> Choose one picture for a particular reasoning move.

01. Embeddings and neighbourhoods: the case



Follow the labelled picture with the matching explanation. Open the vector diagram

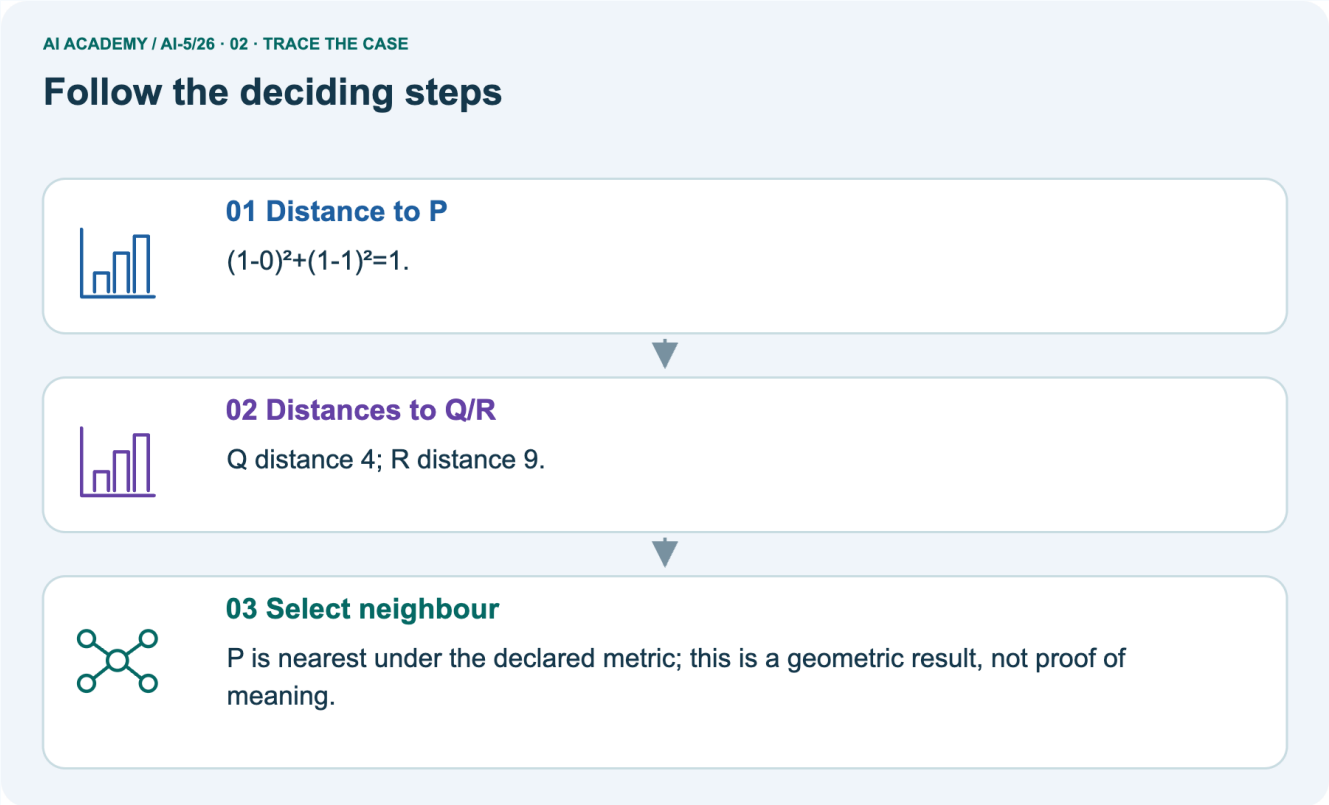
Measure embedding neighbours without turning numerical proximity into factual certainty. Supplied facts: Toy table P =[0, 1], Q=[3, 1], R=[1, 4]; query [1, 1]. Use **squared Euclidean distance**.

All **input**s and outcomes in this worked example are invented classroom data; no real model run is claimed.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



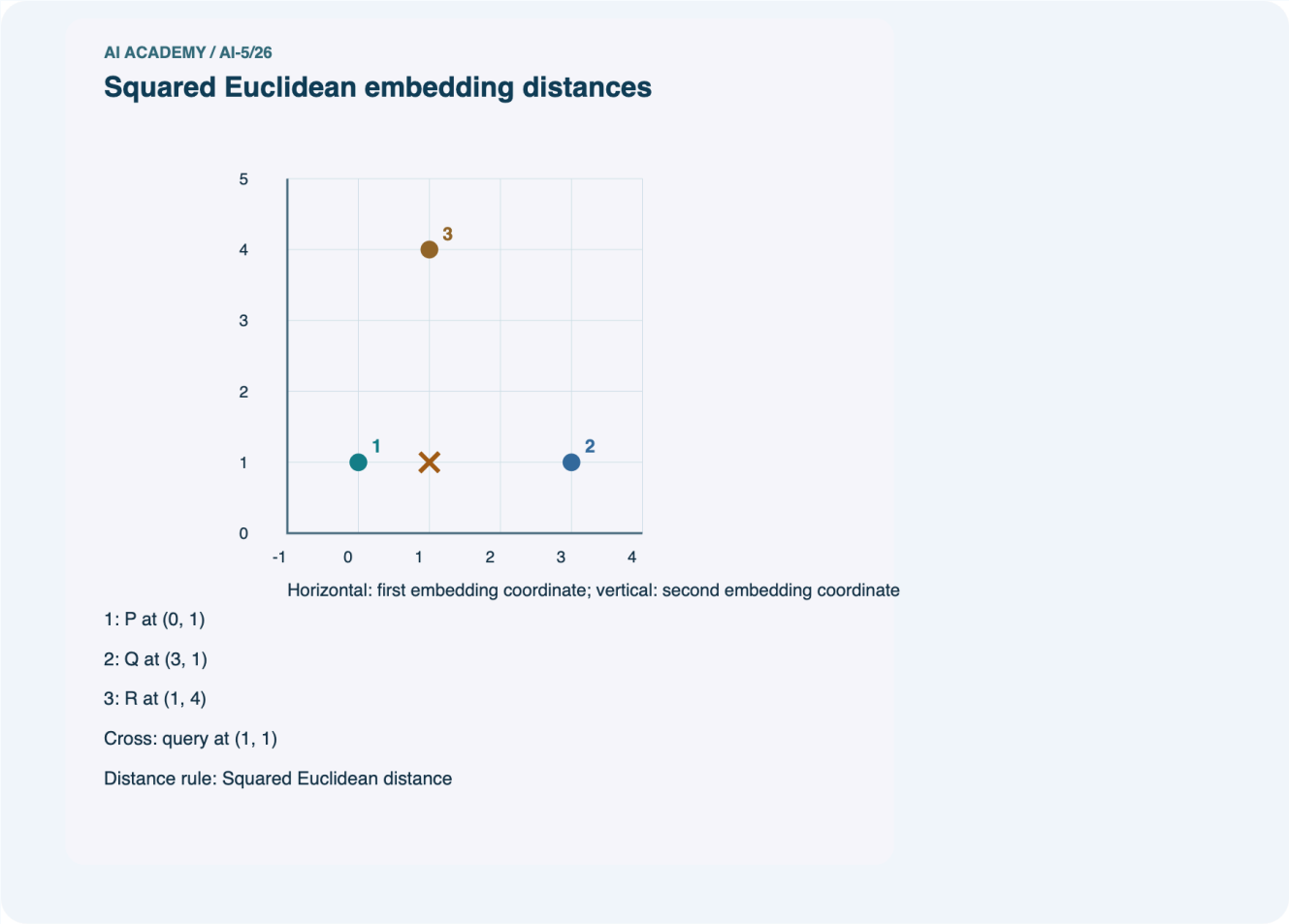
Follow the labelled picture with the matching explanation. Open the vector diagram

Distance to P: (1-0)²+(1-1)²=1. → Distances to Q/R: Q distance 4; R distance 9. → Select neighbour: P is nearest under the declared metric; this is a geometric result, not proof of meaning.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. Squared Euclidean embedding distances



Follow the labelled picture with the matching explanation. Open the vector diagram

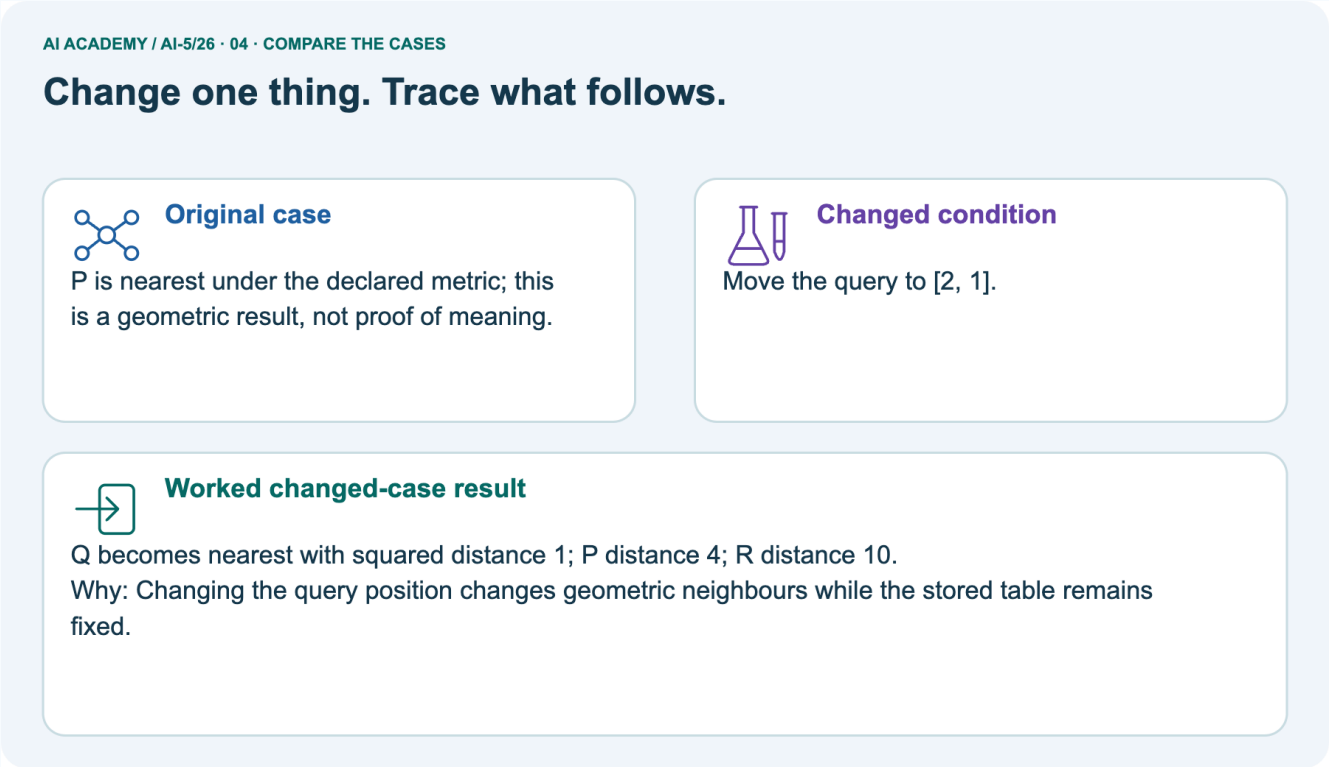
Squared Euclidean embedding distances. P: (0, 1). Q: (3, 1). R: (1, 4). Query query: (1, 1). Axes: first embedding coordinate; second embedding coordinate. Distance: **Squared Euclidean distance**.

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Check row/column meanings, units and denominators before discussing a difference. Supplied toy values are not real class or model results.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: P is nearest under the declared metric; this is a geometric result, not proof of meaning. Changed condition: Move the query to [2, 1].

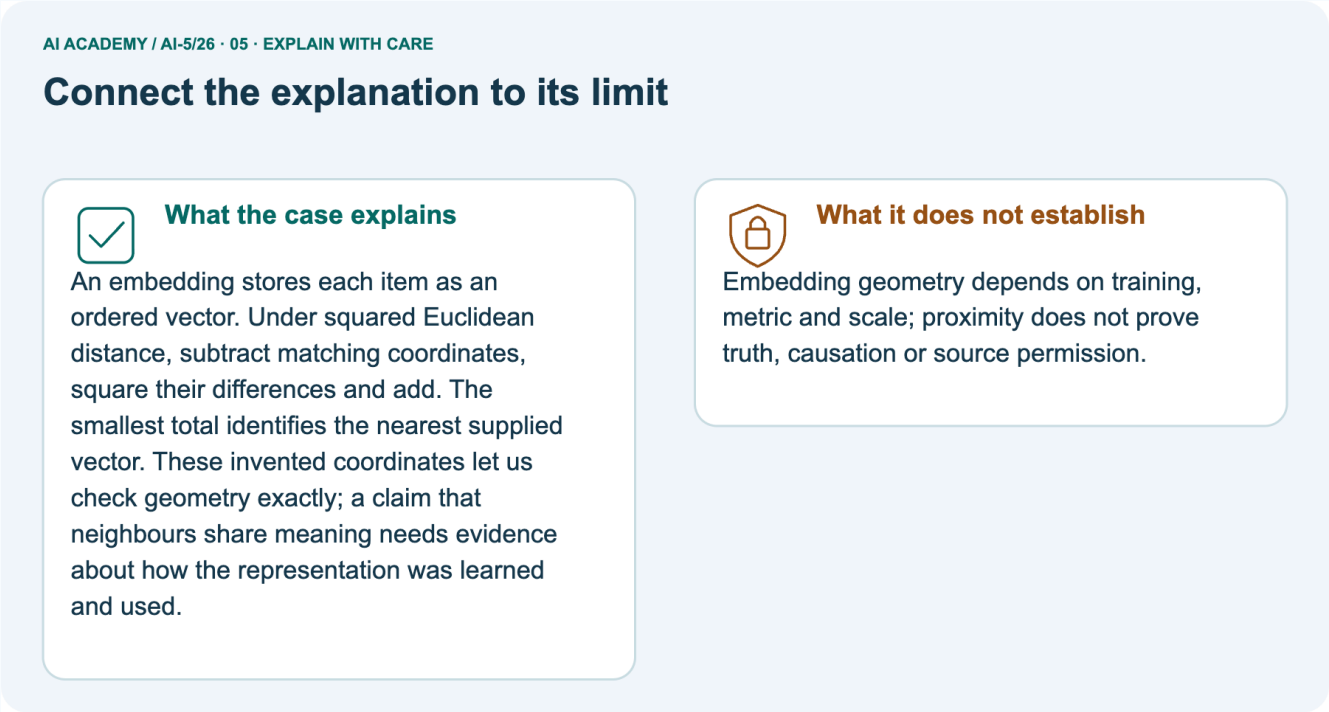
Expected result: Q becomes nearest with squared distance 1; P distance 4; R distance 10. Reason: Changing the query position changes geometric neighbours while the stored table remains fixed.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: Changing the query position changes geometric neighbours while the stored table remains fixed.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

An embedding stores each item as an ordered vector. Under **squared Euclidean distance**, subtract matching coordinates, square their differences and add. The smallest total identifies the nearest supplied vector.

These invented coordinates let us check geometry exactly; a claim that neighbours share meaning needs **evidence** about how the representation was learned and used.

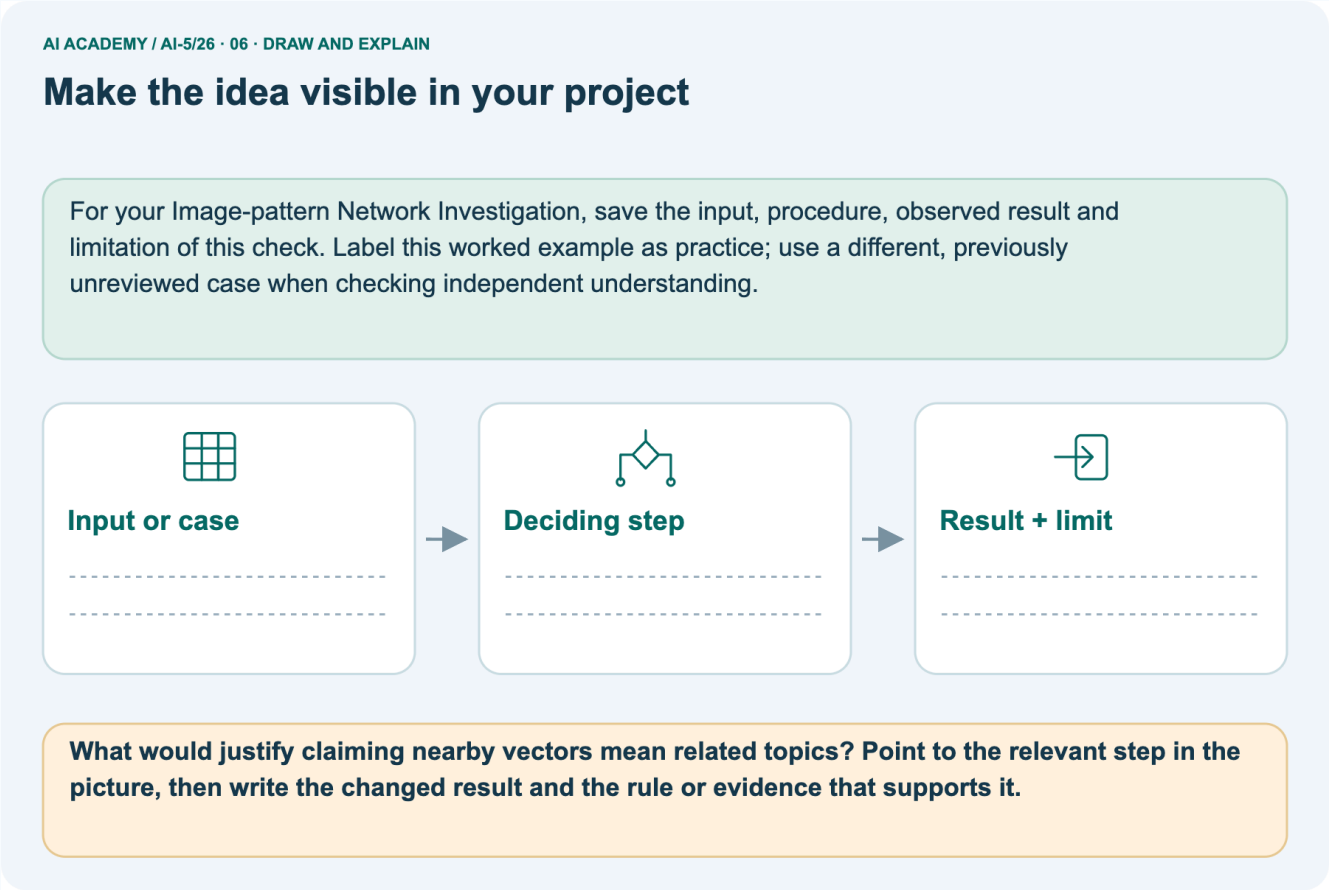
Limit: Embedding geometry depends on **training**, metric and scale; proximity does not prove truth, causation or source permission.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Embedding geometry depends on **training**, metric and scale; proximity does not prove truth, causation or source permission.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For your Image-pattern Network Investigation, save the **input**, procedure, observed result and limitation of this check.

Label this worked example as practice; use a different, previously unreviewed case when checking independent understanding. Draw your **input**, deciding step, result and limit.

Supported question: What would justify claiming nearby vectors mean related topics? Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: **Evidence** about how the table was learned plus task-specific evaluation. These invented coordinates alone establish only distances.

Changed-case check: Q becomes nearest with squared distance 1; P distance 4; R distance 10. Changing the query position changes geometric neighbours while the stored table remains fixed.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Embeddings and neighbourhoods: the case

02. Follow the deciding steps

03. Squared Euclidean embedding distances

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.